

CLASSROOM EXERCISE

Introduction to Symmetrical Three-Phase Systems

Observe the classroom, discuss your ideas and connect everyday electrical loads to a three-phase supply.

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Date and classroom *28.09.2026, F310*

No previous knowledge of three-phase systems is required.

The aim is to start asking useful questions, make observations and build an initial understanding together.

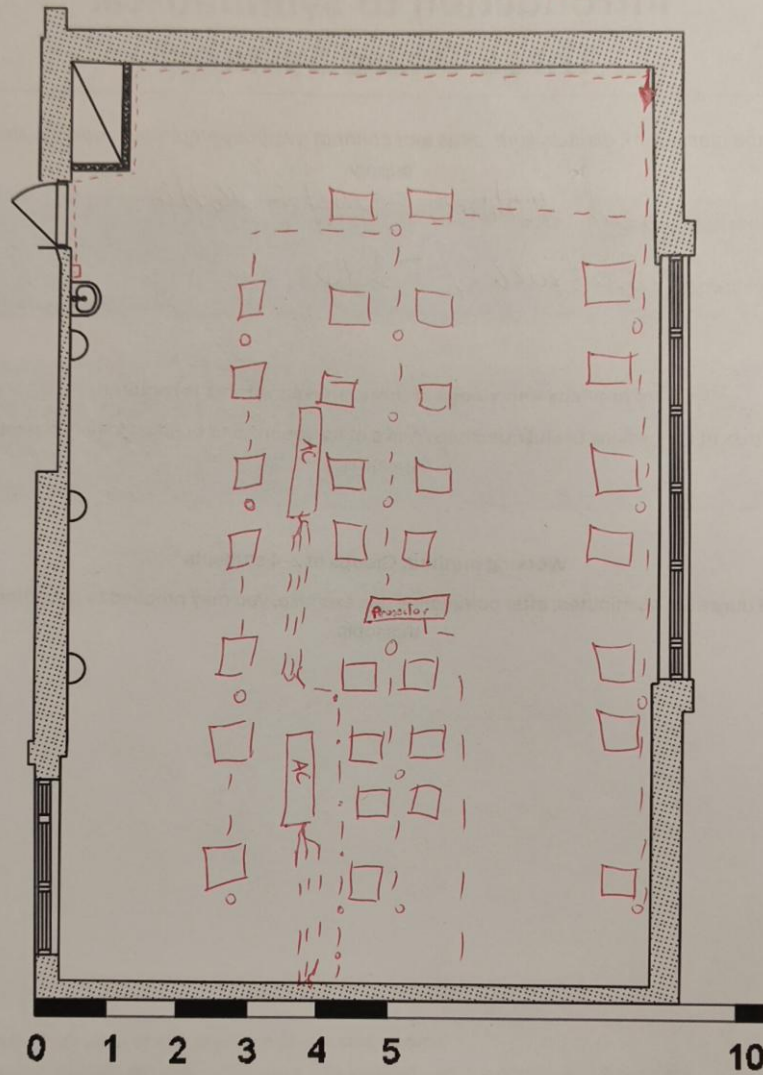
Working method: Groups of 2–4 students

Suggested duration: 45 minutes, after completing this exercise, you may proceed to the other activities in this topic.

jamk

Classroom Floor Plan

Use this floor plan throughout the exercise. Mark the locations of electrical loads and add short notes, symbols or arrows. Use different colours if available.



Floor plan provided for the classroom exercise.

□ Electrical load

○ Outlet or connection point

→ Assumed supply route

Objective and Instructions

The purpose of this exercise is to become familiar with the idea of a symmetrical three-phase system by observing the electrical installations and equipment in the classroom.

After completing the exercise, you should be able to:

- identify common electrical loads in a classroom;
- distinguish between likely single-phase and three-phase loads;
- describe, at an introductory level, what balanced or symmetrical loading means.

Safety and working method

- Observe only. Do not open electrical enclosures, remove covers or disconnect equipment.
- Do not assume that the visible outlet or device reveals the complete electrical installation. Mark uncertain answers as assumptions.
- Record observations on the floor plan and discuss all answers within the group.

Part 1: Identify Electrical Loads

Inspect the classroom and locate as many electrical loads as possible. Examples may include power outlets, lighting, a projector, speakers, ventilation equipment, computers, displays and chargers.

1. Mark the location of each observed load on the floor plan. *→ AC → Compressor, ...*
2. Which loads appear likely to consume the most power? Explain your reasoning.
3. Which loads may operate continuously? *→ AC, Ventilation*
4. Which loads can classroom users control directly? *→ Lights, PC*

Part 2: Single-Phase or Three-Phase?

Discuss whether each load is likely supplied as a single-phase load or as part of a three-phase system. If you are unsure, write "uncertain" and explain what additional information would be needed.

Electrical load	Likely single-phase	Likely three-phase	Reason or evidence	Estimated Total Power (W)
Power outlets <i>IT / Normal</i>	☒	☐		
Lighting	☒	☐		
Projector	☒	☐		
Speakers	☒	☐		
Ventilation system	☐	☒		
Other: <i>Plants</i> <i>Air Conditioning</i>	☒	☒		
Other: <i>Emergency Exit</i>	☐	☐		

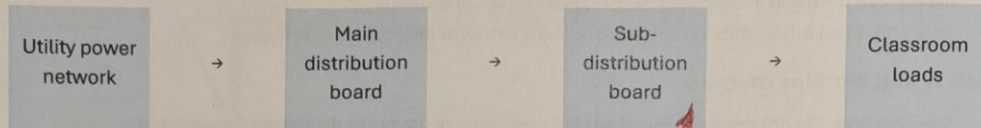
Discuss:

- Why are ordinary wall outlets typically single-phase? *→ because only 230V is required and not much current*
- Why might ventilation equipment use a three-phase motor? *→ because more power is needed and the 3 phase system is perfect for motors*

- Why are high-power motors commonly supplied from a three-phase system

Part 3: Hidden Three-Phase Power

Many classroom devices are single-phase loads, even though the building is supplied by a three-phase electrical system. Complete the supply route below and add it to the floor plan.



5. At what point are individual single-phase circuits formed?

6. Where in the building are all three phases likely to be used?

7. Why is a three-phase supply useful in a large building?

→ until subdistribution in buildings this size

→ ~~to keep the load symmetrical~~

to keep the load symmetrical

Part 4: Balanced Loading

Assume that the classroom lighting is divided into three identical groups. Each group consumes 500 W and each group is connected to a different phase.

L1	L2	L3
500 W	500 W	500 W

8. Would you describe this loading as balanced or symmetrical? Why? *→ yes → all 3 take the same power*
9. What advantages might balanced loading provide? *→ relieves electrical grid*
10. What do you expect to happen to the neutral current when three equal sinusoidal phase currents are displaced by 120° ? *→ 0 A at neutral wire*

Part 5: Unbalanced Loading

Now consider the following load distribution:

L1	L2	L3
1000 W	200 W	300 W

11. Is the loading symmetrical? Explain your answer. *→ no → all 3 have different currents*
12. What practical problems might uneven phase loading cause? *→ bad for power grid*
13. Suggest a better distribution of these loads between L1, L2 and L3 (next page). *→ some wires might have work on their limit (no spare for additional consumer) and some don't have to do nothing*
→ balance it

How would your group distribute the electrical loads evenly among the three phases in this room?

Use the space below to design the distribution.

3 phase direct: AC

1st column PCs: phase 1

2nd column PCs: phase 2

3rd column PCs: phase 3

Lights: phase 1

Clock, Beamer, Speakers, Emergency Exit sign, additional outlet
↳ phase 2

Self-check

- ☒ I can identify examples of electrical loads in a classroom.
- ☒ I can explain the basic difference between a likely single-phase and three-phase load.
- ☒ I can describe what balanced phase loading means.
- ☒ I can now continue to next exercises of this topic.